

Forough Mahpouya (she/her)

Ph.D. Candidate in Industrial and Systems Engineering

Department of Industrial and Systems Engineering, University at Buffalo, The State University of New York

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RESEARCH AREAS

Causal inference and balancing representation learning for AI-enabled decision systems; longitudinal health systems; continuous interventions and time-to-event outcomes; measurement, fairness, and treatment-effect heterogeneity; sequential experimentation and manufacturing systems.

EDUCATION

Ph.D. Candidate, Industrial and Systems Engineering 2022–present

University at Buffalo, The State University of New York, Buffalo, NY

Dissertation: Quantitative Modeling of Health Outcomes in Patient–Caregiver Dyads in End-of-Life Care

Advisors: Alexander Nikolaev and Sabrina Casucci

M.Sc., Industrial Engineering 2017–2019

K. N. Toosi University of Technology, Tehran, Iran

Thesis: A Bootstrap Approach to Introduce a Robust Multivariate Control Chart

Advisors: Hamid Shahriari and Emad Roghanian

B.Sc., Industrial Engineering 2006–2010

Iran University of Science and Technology, Tehran, Iran

RESEARCH EXPERIENCE

Doctoral Researcher 2023–present

Department of Industrial and Systems Engineering, University at Buffalo, Buffalo, NY

- **Balancing representations for continuous interventions and time-to-event outcomes.** Lead theory development of a neural architecture for conditional treatment effects of continuous-dose interventions on survival outcomes; derive identification conditions and link the objective to hazard modeling.
- Curate and analyze observational caregiving data; design and implement algorithms; and develop technical manuscripts and supporting theory, including proofs.
- **Heterogeneous treatment effects and subgroup identification.** Developed a taxonomy of model-based partitioning, uplift and loss-based learners, causal trees and forests, and representation learning methods, plus a decision map connecting data structure and clinical objective with stability, interpretability, and fairness.
- **Fairness in glomerular filtration rate prediction.** Evaluated adversarial balancing-based representation learning for fairness-aware GFR prediction and co-authored an NIDDK data request specifying the rationale and required biomarkers.

Researcher, Anomaly Detection in Inkjet Printing 2022–2023

Department of Industrial and Systems Engineering, University at Buffalo, Buffalo, NY

- Led theory development and preliminary designs for classification-based anomaly detection of inkjet-printing droplets through analysis of pinch-off behavior; documented results, identified gaps, and proposed methodological improvements.

Research Collaborator 2017–2022

Department of Industrial Engineering, K. N. Toosi University of Technology, Tehran, Iran

- Organized literature-review manuscripts on mean-vector estimation for multivariate populations with outliers and on degradation-data models for product quality and stability in the food industry.

PUBLICATIONS AND MANUSCRIPTS

Refereed Journal Articles

Mahpouya, F., Burris, C. J., Paul, H., & Nikolaev, A. (2026). Maximizing the expected value of experimentation for finding top- κ rank via aggregation of pairwise comparisons. *IIE Transactions*. <https://doi.org/10.1080/24725854.2026.2645062>

Mahpouya, F., Shahriari, H., & Roghanian, E. (2022). Design of a robust T^2 control chart: A re-sampling approach. *Quality and Reliability Engineering International*, 38(2), 924–940. <https://doi.org/10.1002/qre.3022>

Manuscripts Under Review or Revision

Das, S., Casucci, S., **Mahpouya, F.**, & Nikolaev, A. (2026). Treatment effect estimation from observational data in healthcare: Study designs, inherent biases, and a propensity score-focused quality assessment scale. Manuscript under revision for *Socio-Economic Planning Sciences*.

Mahpouya, F., Das, S., Nikolaev, A., & Casucci, S. (2026). Use of observational causal inference in dementia management studies: A systematic literature review. Manuscript under revision for *PLOS Digital Health*.

Walker, R. J., Linde, S., Egede, L. E., **Mahpouya, F.,** Nikolaev, A., Fraser, R., & Martinez, J. (2026). Assessing the Healthy Eating Index and health: Psychometric properties and prediction of cardiometabolic outcomes. Manuscript under review at *Hypertension*.

Works in Progress

Mahpouya, F., Nikolaev, A., & Casucci, S. (2026). Adversarial balancing for estimating the causal effect of caregiver burden on patient long-term-care transitions. Manuscript in preparation.

Mahpouya, F., Casucci, S., & Sullivan, S. (2026). A population-based, bidirectional measure of caregiver risk and resilience: The Caregiver Burden–Benefit Index (CareBI). Manuscript in preparation.

Rogers, A. H., Fligelman, S. J., Pester, B. D., Pielech, M., Klages, K. L., **Mahpouya, F.,** Nikolaev, A., Groenewald, C. B., Ketterl, T. G., Bevers, L. M., Dietel, K. E., & Palermo, T. M. (2026). Pain management and opioid use in adolescents and young adults with cancer: A qualitative analysis of patient and caregiver experiences. Manuscript in preparation.

CONFERENCE PRESENTATIONS

Mahpouya, F., Nikolaev, A., & Casucci, S. (2026). Adversarial balancing for estimating the causal effect of caregiver burden on patient long-term-care transitions. [Conference presentation]. *IISE Annual Conference & Expo*, Arlington, TX.

Mahpouya, F., Casucci, S., Sullivan, S., & Barrick, C. (2025). Representing caregiver burden in observational studies: Development of the Caregiver Burden Index (CareBI) using NSOC. [Conference presentation]. *INFORMS Annual Meeting*, Atlanta, GA.

Mahpouya, F., Shahriari, H., & Roghanian, E. (2020). Use of resampling techniques to design a robust Hotelling's T^2 control chart for limited data. [Conference presentation]. *16th Iran International Industrial Engineering Conference*, Alzahra University, Tehran, Iran. Conference record

Mahpouya, F., & Shahriari, H. (2019). A bootstrapping approach to introduce a robust multivariate control chart. [Conference presentation]. *15th Iran International Industrial Engineering Conference*, Yazd University, Yazd, Iran. Conference record

TEACHING EXPERIENCE

Teaching Assistant and Guest Lecturer 2022–2026
University at Buffalo, Buffalo, NY

Course areas: simulation and stochastic models; production planning and control; applied probability and statistics; healthcare systems engineering; computer-integrated manufacturing; social-network behavior models.

IE 505: Production Planning and Control Spring 2026

IE 477: Simulation Modeling and Analysis Fall 2023, 2024, 2025

- Led 12 recitation sessions and two course lectures each semester; designed six weeks of recitation material using @RISK and Arena.

IE 406/506: Computer-Integrated Manufacturing Fall 2023–Fall 2025

- Led four lab tutorials each semester using Autodesk Fusion 360 for 3D CAD modeling and manufacturing.

IE 551: Simulation and Stochastic Models Spring 2024, 2025

- Delivered a course lecture and held group-specific active office hours.

IE 511: Social Network Behavior Models Spring 2024, 2025

IE 585: Industrial Engineering Principles and Practice in Healthcare Fall 2024

- Delivered a guest lecture on PICO question formulation and analysis of complex observational healthcare data for interpretable evidence.

EAS 305: Applied Probability and Statistics Fall 2022–Summer 2023

- Led 54 recitations in Fall 2022 and 30 in Spring 2023; designed ten weeks of recitation material; delivered a Minitab lab and an exam-review lecture each semester.

INDUSTRY AND PROFESSIONAL EXPERIENCE

Industrial Engineer / Systems Optimizer

2012–2016

Arioplast, photo-frame profile manufacturing, Rasht, Iran

- Reorganized operations and workflows, freeing two employees to staff two newly added production lines.
- Co-designed an automated weighing and pricing system that categorized products, monitored manufacturing waste in real time, and enabled immediate pricing of customer orders.
- Developed process instructions and operating manuals and trained staff on the new procedures.

Part-Time Technical Mentor

2010–2012; 2015–2017

Private training institutes, Rasht and Sari, Iran

- Designed and delivered tutorials and mentored students in ICDL, Visio, and MATLAB.

HONORS AND AWARDS

- **University at Buffalo ISE Graduate Student Teaching Award**, 2025–2026.
- **IISE Future Faculty Fellows (3F) Program**, selected fellow, 2025–2026.
- **Exceptionally Talented Student designation**, M.Sc. program, K. N. Toosi University of Technology, 2021.
- Ranked **3rd of 36** students in the M.Sc. program, K. N. Toosi University of Technology, 2019.
- Ranked in the **top 1% of 15,800 participants** in Iran's national M.Sc. entrance examination in Industrial Engineering, 2017.

PROFESSIONAL AND ACADEMIC SERVICE

- Reviewed two manuscripts for the *Journal of Nursing Measurement*.
- Contributed to the University at Buffalo's *Effective Pedagogy for Teaching Assistants* program by sharing approaches to recitation design and addressing teaching-assistant challenges through Universal Design for Learning, Spring 2025.
- Mentored undergraduate students on university projects and statistical analysis using MATLAB, 2010–2022.

METHODS AND TECHNICAL EXPERTISE

Causal and Machine Learning Methods: Causal inference; balancing and adversarial representation learning; continuous-treatment effect estimation; treatment-effect heterogeneity and subgroup analysis; longitudinal confounding; survival and hazard modeling; neural networks; fairness-aware learning.

Statistical and Systems Methods: Psychometric and bifactor measurement; simulation and stochastic modeling; sequential experimentation and ranking; experimental design; anomaly detection; robust multivariate statistics and process control.

Applications: Aging and caregiving systems; healthcare delivery and end-of-life care; cardiometabolic health; manufacturing quality and process improvement.

Programming and Research Tools: Python; R; MATLAB.

Technical Software: Arena; @RISK; Autodesk Fusion 360; Minitab.